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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 03

Subject:	DevOps Lab
TCODE:	TE7950
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Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	2025 – 2026, Semester VII
Date of Performance	16th July, 2026
Title of Assignment:	Simulating Git Collaboration & Conflict Resolution

Theory:	Git collaboration allows multiple developers to work on the same project simultaneously using separate branches. When different branches modify the same part of a file, Git cannot automatically combine the changes and reports a merge conflict. The conflict must be resolved manually by editing the affected file, removing the conflict markers, and selecting the final version of the content. After resolving the conflict, the changes are staged, committed, and pushed to the remote repository. This process ensures that all changes are correctly integrated while maintaining the project's history. **Commands Used** 1. git checkout main – Switched to the main branch before starting the experiment. 2. git status – Verified that the working tree was clean and the main branch was up to date. 3. git checkout -b conflicting-feature – Created a new branch named conflicting-feature and switched to it. 4. Modified the README.md file and changed the first line on the conflicting-feature branch. 5. git add README.md – Staged the modified README.md file on the feature branch.

	6. git commit -m "Update README first line on feature branch" – Committed the changes made on the conflicting-feature branch. 7. git checkout main – Switched back to the main branch. 8. Modified the same first line of the README.md file differently on the main branch. 9. git add README.md – Staged the modified file on the main branch. 10. git commit -m "Update README first line on main" – Committed the changes on the main branch. 11. git merge conflicting-feature – Attempted to merge the conflicting-feature branch into the main branch, resulting in a merge conflict. 12. git status – Verified the conflict by checking that README.md appeared under Unmerged paths. 13. Manually edited the README.md file to remove the conflict markers and resolve the conflict. 14. git add README.md – Staged the resolved file after resolving the merge conflict. 15. git status – Confirmed that all conflicts were fixed and the repository was ready for the merge commit. 16. git commit -m "Merge conflicting-feature, resolve README conflict" – Completed the merge by creating a merge commit after resolving the conflict. 17. git push origin main – Pushed the resolved merge commit from the local main branch to the remote GitHub repository.

Implementation:

Step 1: Verify Current Repository Status

```
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % mkdir -p DevOps-Assignments/Assignment-03-Collaboration

cat > DevOps-Assignments/Assignment-03-Collaboration/README.md <<'EOF'
# Git Collaboration Demo

This is the original content on the main branch.
EOF

git add DevOps-Assignments/Assignment-03-Collaboration/README.md
git commit -m "Add Assignment 3 conflict demo"
[main 3f2a0d3] Add Assignment 3 conflict demo
 1 file changed, 3 insertions(+)
 create mode 100644 DevOps-Assignments/Assignment-03-Collaboration/README.md
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git checkout -b conflicting-feature
Switched to a new branch 'conflicting-feature'
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % cat > DevOps-Assignments/Assignment-03-Collaboration/README.md <<'EOF
# Git Collaboration Demo - Feature Branch Version

This change was made on the conflicting-feature branch.
EOF

git add DevOps-Assignments/Assignment-03-Collaboration/README.md
git commit -m "Update README on conflicting feature branch"
[conflicting-feature df5e7b7] Update README on conflicting feature branch
 1 file changed, 2 insertions(+), 2 deletions(-)
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)
```

Verified Repository Status — Confirmed that the repository was on the main branch with a clean working tree before creating the conflicting branch.

Step 2: Create and Switch to a New Branch

```
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git checkout main
Already on 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps %
```

Created Conflicting Branch — Created a new branch named conflicting-feature from the current state of the main branch.

Step 3: Modify README.md on the Feature Branch

```
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % cat > DevOps-Assignments/Assignment-03-Collaboration/README.md <<'EO
# Git Collaboration Demo

This is the original content shared by both branches.
EOF

git add DevOps-Assignments/Assignment-03-Collaboration/README.md
git commit -m "Reset Assignment 3 conflict demo baseline"
[main 90406e1] Reset Assignment 3 conflict demo baseline
 1 file changed, 2 insertions(+), 2 deletions(-)
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git sw
Switched to a new branch 'assignment-03-feature'
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % cat > DevOps-Assignments/Assignment-03-Collaboration/README.md <<'EO
# Git Collaboration Demo - Feature Branch

This version was written by Developer A.
EOF

git add DevOps-Assignments/Assignment-03-Collaboration/README.md
git commit -m "Update README on Assignment 3 feature branch"
[assignment-03-feature 70e6404] Update README on Assignment 3 feature branch
 1 file changed, 2 insertions(+), 2 deletions(-)
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git log --oneline -3
cat DevOps-Assignments/Assignment-03-Collaboration/README.md
70e6404 (HEAD -> assignment-03-feature) Update README on Assignment 3 feature branch
90406e1 (main) Reset Assignment 3 conflict demo baseline
df5e7b7 (conflicting-feature) Update README on conflicting feature branch
# Git Collaboration Demo - Feature Branch

This version was written by Developer A.
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps %
```

Open file in editor (cmd + click)

Updated Feature Branch — Modified the first line of README.md on the conflicting-feature branch and committed the changes.

Step 4: Switch Back to Main and Make a Different Change

```
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git merge conflicting-feature
Updating 3f2a0d3..df5e7b7
Fast-forward
 DevOps-Assignments/Assignment-03-Collaboration/README.md | 4 ++--
 1 file changed, 2 insertions(+), 2 deletions(-)
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git status
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % cat > DevOps-Assignments/Assignment-03-Collaboration/README.md <<'E
# Git Collaboration Demo - Conflict Resolved

This assignment demonstrates resolving conflicting changes from
the main and conflicting-feature branches.

The final version combines the purpose of both changes.
EOF
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % cat DevOps-Assignments/Assignment-03-Collaboration/README.md
# Git Collaboration Demo - Conflict Resolved

This assignment demonstrates resolving conflicting changes from
the main and conflicting-feature branches.

The final version combines the purpose of both changes.
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps %
```

Updated Main Branch — Switched back to the main branch, modified the same line differently, and committed the changes.

Step 5: Merge the Branch

```
 1 file changed, 2 insertions(+), 2 deletions(-)
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git log --oneline --graph --all -6
cat DevOps-Assignments/Assignment-03-Collaboration/README.md
* 11ea06b (HEAD -> main) Update README on Assignment 3 main branch
| * 70e6404 (assignment-03-feature) Update README on Assignment 3 feature branch
|/
* 90406e1 Reset Assignment 3 conflict demo baseline
* df5e7b7 (conflicting-feature) Update README on conflicting feature branch
* 3f2a0d3 Add Assignment 3 conflict demo
* 64d5604 (origin/main, origin/HEAD, feature-update) Update README on feature branch
# Git Collaboration Demo - Main Branch

This version was written by Developer B.
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git merge assignment-03-feature
Auto-merging DevOps-Assignments/Assignment-03-Collaboration/README.md
CONFLICT (content): Merge conflict in DevOps-Assignments/Assignment-03-Collaboration/README.md
Automatic merge failed; fix conflicts and then commit the result.
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps %
```

Merge Conflict — Attempted to merge the conflicting-feature branch into main, resulting in a merge conflict because both branches modified the same

line. **Step 6: Verify the Conflict**

```
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git status
On branch main
Your branch is ahead of 'origin/main' by 4 commits.
  (use "git push" to publish your local commits)

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   DevOps-Assignments/Assignment-03-Collaboration/README.md

no changes added to commit (use "git add" and/or "git commit -a")
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps %
```

Conflict Verified — Confirmed the merge conflict using git status, which displayed README.md under Unmerged paths.

Step 7: Resolve the Conflict

Conflict Resolved — Manually removed the conflict markers and edited the file to create the final merged version.

Step 8: Stage the Resolved File

```
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % cat DevOps-Assignments/Assignment-03-Collaboration/README.md
<<<<<<< HEAD
# Git Collaboration Demo - Main Branch

This version was written by Developer B.
=======
# Git Collaboration Demo - Feature Branch

This version was written by Developer A.
>>>>>>> assignment-03-feature
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % cat > DevOps-Assignments/Assignment-03-Collaboration/README.md <<
# Git Collaboration Demo - Conflict Resolved

This assignment demonstrates collaboration between Developer A
on the feature branch and Developer B on the main branch.

Both branches modified the same file, resulting in a merge conflict.
The conflict was resolved by combining the purpose of both changes.
EOF
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % cat DevOps-Assignments/Assignment-03-Collaboration/README.md
# Git Collaboration Demo - Conflict Resolved

This assignment demonstrates collaboration between Developer A
on the feature branch and Developer B on the main branch.

Both branches modified the same file, resulting in a merge conflict.
The conflict was resolved by combining the purpose of both changes.
```

Resolved File Staged — Staged the resolved file and confirmed that all merge conflicts had been fixed.

Step 9: Complete the Merge

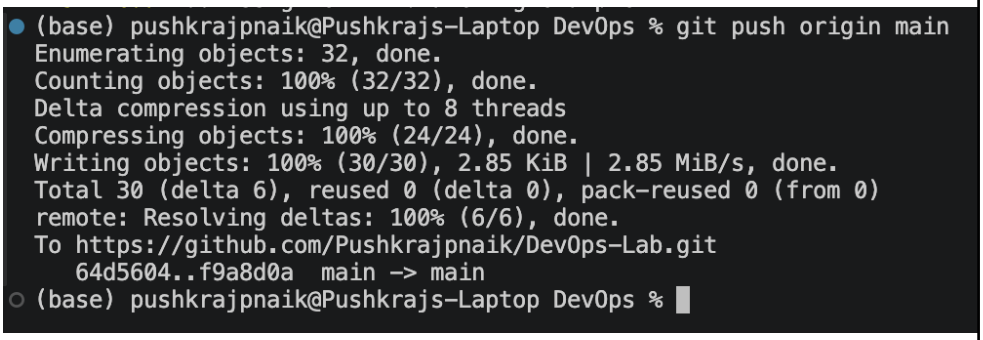
```
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git add DevOps-Assignments/Assignment-03-Collaboration/README.md
git status
On branch main
Your branch is ahead of 'origin/main' by 4 commits.
  (use "git push" to publish your local commits)

All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
        modified:   DevOps-Assignments/Assignment-03-Collaboration/README.md

(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git commit -m "Resolve Assignment 3 README merge conflict"
[main f9a8d0a] Resolve Assignment 3 README merge conflict
(base) pushkrajpnaik@Pushkrajs-Laptop DevOps % git log --oneline --graph --all -8
*   f9a8d0a (HEAD -> main) Resolve Assignment 3 README merge conflict
|\
| * 70e6404 (assignment-03-feature) Update README on Assignment 3 feature branch
* | 11ea06b Update README on Assignment 3 main branch
|/
* 90406e1 Reset Assignment 3 conflict demo baseline
* df5e7b7 (conflicting-feature) Update README on conflicting feature branch
* 3f2a0d3 Add Assignment 3 conflict demo
* 64d5604 (origin/main, origin/HEAD, feature-update) Update README on feature branch
* 577feda Add Assignment 2 branching example
```

Merge Completed — Completed the merge by creating a merge commit that combines the histories of the main and conflicting-feature branches after

	resolving the conflict. **Step 10: Push to GitHub**  Pushed to GitHub — Uploaded the resolved merge commit to the remote GitHub repository.

Subject Faculty Name: Prof. Arijit Dutta